SpendWise: A Simple Backend Learning Project

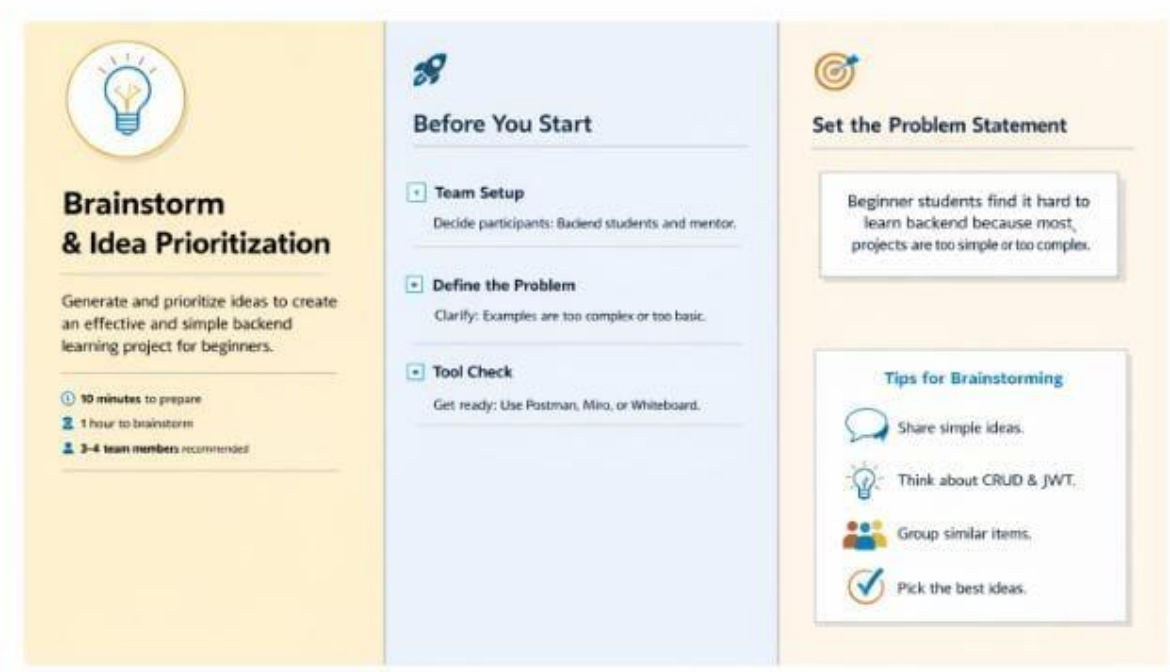
Ideation Phase
Brainstorm & Idea Prioritization

Date	03 March 2026
Team ID	NM2025TMID40009
Project Name	SpendWise
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming creates an open environment where team members can freely share ideas to solve a problem. For SpendWise, the goal is to generate simple, practical ideas that help students learn backend development. All ideas are first listed without judging, then grouped, and finally prioritized based on usefulness and feasibility for beginners.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping

2

Brainstorm

Write down any ideas that come to mind to support the backend learning project.

10 minutes

TIP

Think about what you struggled to understand. Share ideas that keep the project simple, focused, and easy to learn.

Person 1	Person 2	Person 3	Person 4
Implement Register & Login with JWT authentication.	CRUD API for adding, viewing, updating, and deleting user expenses.	Use MongoDB with Mongoose to store users and expenses.	Simple and clear folder structure with routes, controllers, and models.

3

Group ideas

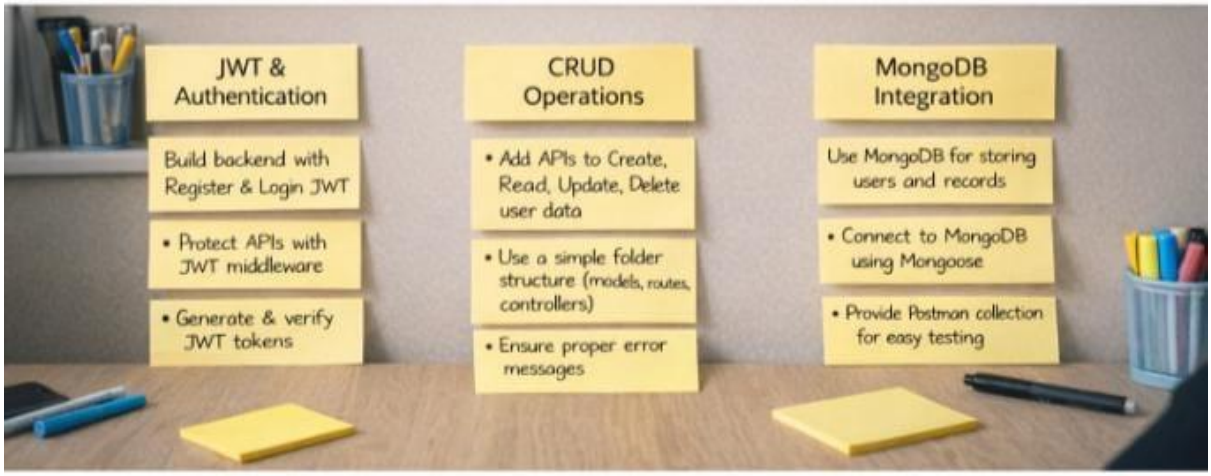
Take turns sharing youdeas and group similar sticky notes into useful themes for **SpendWise**. Give each group a clear label describing the theme once all sticky notes are clustered.

20 minutes

TIP

Add tags to sticky notes to find, organize, and categorize ideas.

- Key tags to consider:
  - CRUD Operations
  - JWT & Authentication
  - MongoDB Integration



Step-3: Idea Prioritization

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	API testing interface or optional web UI	Postman / Thunder Client / (Optional) React
2.	Application Logic	Handles authentication, CRUD operations, and business logic	Node.js, Express.js
3.	Database Layer	Stores users and records/transactions	MongoDB, Mongoose
4.	Authentication	Secures APIs using token-based authentication	JWT (JSON Web Token), bcrypt
5.	Configuration	Environment variables and DB connection setup	dotenv, Node.js
6.	API Layer	Exposes REST APIs for register, login, and CRUD operations	Express.js, REST APIs

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Uses open-source backend frameworks and libraries	Node.js, Express.js, Mongoose
2.	Secure Authentication	Protects routes using token-based authentication	JWT, bcrypt

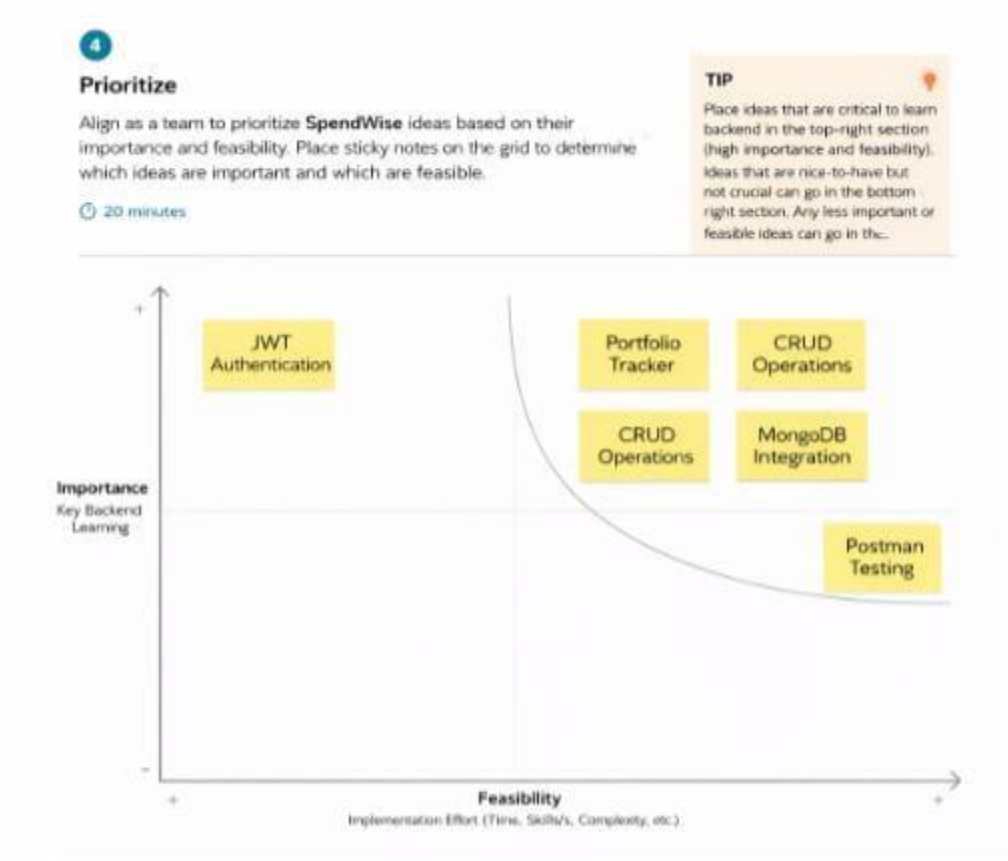
3. Database Integration Connects and performs operations on NoSQL database MongoDB

4. RESTful Architecture Follows REST principles for API design HTTP, Express.js

5. Scalable Structure

Organized folder structure for controllers, models, routes, middleware Node.js Project Structure

6. Environment Configuration Keeps secrets and config outside code dotenv (.env file)

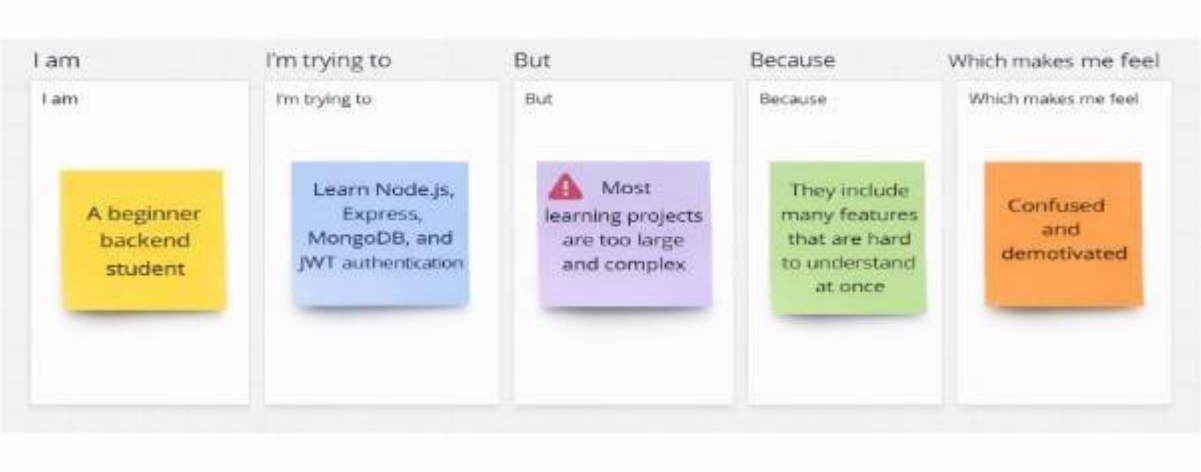


Customer Problem Statement Template:

A customer problem statement helps understand the user's point of view while learning from a product. In the SpendWise project, it focuses on the needs of beginner backend students who want to learn Node.js, Express, MongoDB, and JWT in a simple way. It helps identify problems like confusion with complex projects and difficulty connecting concepts, so the solution stays clear, useful, and easy to learn.

I am	A beginner backend student
I'm trying to	Learn Node.js, Express, MongoDB, and JWT authentication
but	Most learning projects are too large and complex
because	They include many features that are hard to understand at once
which makes me feel	Confused and demotivated
	Confused and demotivated

Example:



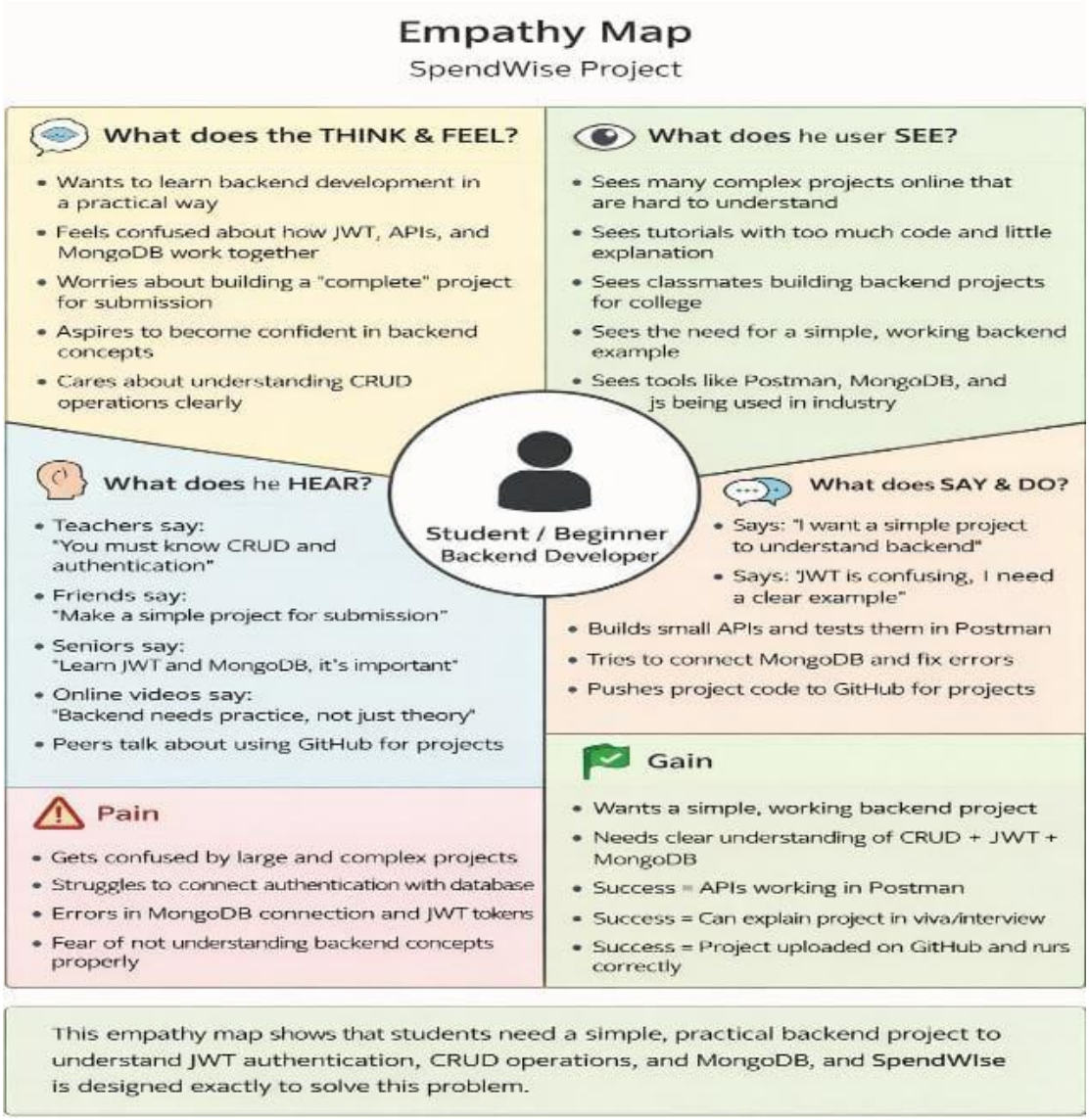
Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A beginner backend student	Learn Node.js, Express, MongoDB, and JWT authentication	Most learning projects are too large and complex	They include many features that are hard to understand at once	Confused and demotivated
PS-2	A college student	Build a simple backend project for practice and submission	I don't find a clear example focused only on CRUD and authentication	Tutorials often jump to advanced topics without explaining basics	Unsure about my understanding
PS-3	A beginner developer	Connect JWT authentication with MongoDB and APIs	I keep getting errors in tokens or database connection	The flow between login, token, and protected routes is not clear	Frustrated and stuck

PS-4	A student preparing for viva or interviews	Explain CRUD operations and JWT authentication with a project	I don't have a simple end-to-end backend example	Most available projects are either too basic or too advanced	Nervous and underprepared

Empathy Map Canvas:

An empathy map is a simple way to understand a beginner's thoughts, feelings, and actions while learning backend development. It helps identify how beginners work with APIs, JWT authentication, and MongoDB, and what problems they face with complex code. Creating this empathy map allows the project to be viewed from the beginner's perspective, ensuring the solution stays simple, practical, and easy to understand.

Example:



Says

- I want a simple project to learn backend
- I understand theory, but coding it is different
- JWT is confusing, I need a clear example
- I just want to register, login, and manage data

What does the user SEE?

- How do APIs, JWT, and MongoDB connect together?
- This project should not be too complex
- I need practice for interviews
- I hope I can remember this for my viva

Student / Beginner
Backend Developer

Does

- Searches for simple Node.js + MongoDB projects
- Watches tutorials and tries them out
- Tests APIs in Postman/Thunder Client
- Pushes code to GitHub for practice

Feels

- Confused by large projects with lots of code
- Happy when small APIs work as expected
- Frustrated when JWT tokens do not work
- Motivated if I can explain my backend in in a viva

Pain

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Feels

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